14. Steps to sketch full phase portrait

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Steps for drawing the phase portrait for an autonomous system $\dot{x} = f(x,y), \dot{y} = h(x,y)$:

1. Solve the system

$$f(x,y) = 0$$
$$h(x,y) = 0$$

to find all the critical points in the (x,y)-phase plane. The critical point itself is a trajectory called a stationary trajectory.

2. Calculate the Jacobian matrix

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$$\mathbf{J}(x,y) = \begin{pmatrix} \partial f/\partial x & \partial f/\partial y \\ \partial h/\partial x & \partial h/\partial y \end{pmatrix} = \begin{pmatrix} f_x & f_y \\ h_x & h_y \end{pmatrix}. \quad (4.90)$$

This will be a 2×2 matrix whose entries are functions of x and y.

3. At each critical point (a,b), compute the 2×2 matrix $\mathbf{A} = \mathbf{J}(a,b)$ by evaluating $\mathbf{J}(x,y)$ at (a,b). The linear approximation of the system near (a,b) is then

$$\begin{pmatrix} \dot{X} \\ \dot{Y} \end{pmatrix} \approx \mathbf{A} \begin{pmatrix} X \\ Y \end{pmatrix} \quad \text{where} \quad \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} x-a \\ y-b \end{pmatrix}.$$

Mark the critical point (a,b) in the (x,y)-plane, and draw a miniature copy of the phase portrait of the linear approximation centered at (a,b). Indicate with arrowheads the direction of motion on the trajectories near the critical point.

Warning: This step is justified only if the linearized system is **structurally stable** . That is, if the matrix $\mathbf{A}$ lies in one of the big regions in the trace-determinant plane, or equivalently, if the phase portrait of $\mathbf{A}$ is a saddle, a node, or a spiral. We will discuss this in more detail in the next lecture.

4. Connect trajectories emanating from or approaching critical points, while making sure **trajectories never cross or touch** . Trajectories can appear to meet at critical points (as $t \to \pm\infty$). If there are nearly parallel trajectories which seem to point in opposite directions, then this often means that you made a mistake in one of the previous steps, or you made a drawing error.

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- Very helpful — Useful and practical, refer back to this a lot. (1)
- Boundary Regions — What happens if matrix A is on a boundary region in the Det/Tr plane? (2)

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